

1] Problem Statement:

Customer return behavior on Amazon is shaped by multiple interacting factors, including product attributes, pricing patterns, retailer performance, and the sentiment expressed in customer reviews. However, the relationship between these variables and actual return likelihood remains poorly understood at scale. This limited visibility makes it difficult to identify which products are most frequently returned, why customers choose to return them, and which retailers consistently contribute to return-driven dissatisfaction. As a result, both customers and businesses lack clear insight into the underlying drivers of returns, affecting recommendation quality, product trust, and operational efficiency.

2] Dataset URL :-

<https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-023/tree/main/raw>

3] Objectives :-

- 1] Perform exploratory and descriptive data analysis on Amazon product metadata and review data to identify trends, patterns, and anomalies in ratings, pricing, review volume, and customer behavior across products, categories, and sellers.
- 2] Develop a predictive model that estimates the likelihood of a product being returned using structured metadata, pricing trends, customer ratings, seller information, and review-derived features.
- 3] Extract and categorize return-related reasons from customer reviews using NLP techniques, enabling a deeper understanding of issues such as quality defects, size problems, misleading descriptions, and customer dissatisfaction.
- 4] Quantify and analyze sentiment patterns within reviews to understand how emotional tone and aspect-specific sentiment correlate with return behavior across different product categories.
- 5] Identify high-risk product categories and subcategories by calculating a return risk score based on historical review signals, allowing Amazon to pinpoint categories that require quality improvement.
- 6] Evaluate retailer performance by detecting sellers associated with consistently high return-related complaints, enabling better enforcement of quality standards.

7] Build a quality-aware recommendation system that suggests alternative products with lower return risk and stronger reliability indicators to improve customer satisfaction.

8] Generate business insights that support operational decision-making related to inventory management, seller evaluation, and product quality enhancement.

4] Data Dictionary

1. Amazon fashion

Column Name	Description	Example Value	Datatype
rating	Star rating given by user	5.0 / 4.0	float
title	Review title	"Such a lovely scent..."	string
text	Full review text	"This spray is really nice..."	string
images	List of image URLs (empty in your data)	[]	array/list
asin	Amazon product ID	"B00YQ6X8EO"	string
parent_asin	Parent product ASIN (used for variations)	"B00YQ6X8EO"	string
user_id	Unique reviewer ID	"AGKHLEW2SOWHNMFAQJGBEC7INQ"	string
timestamp	Unix timestamp (ms)	1588687728923	long / int64
helpful_vote	Number of helpful votes	0 / 1	integer
verified_purchase	Whether the review is verified	true / false	boolean

2. Clothing,Shoes,Jewellery

Column Name	Datatype	Description	Example
rating	float	User rating (1–5 stars)	3.0
title	string	Title of the review	"Arrived Damaged : liquid in hub locker!"
text	string	Full review text	"Unfortunately Amazon in their wisdom..."
images	array of objects	Each object contains small/medium/large image URLs + type	[{"small_image_url": "..."}]
asin	string	Product identifier	"B096S6LZV4"
parent_asin	string	Parent ASIN (variation parent)	"B09NSZ5QMF"
user_id	string	Reviewer's unique ID	"AFKZENTNBQ7A7V7UXW5JJI6UGRYQ"
timestamp	long / int64	Review timestamp (Unix ms)	1677938767351
helpful_vote	integer	Number of helpful votes	0
verified_purchase	boolean	True if Amazon verified it	true / false

3. Electronics

Column Name	Datatype	Description	Example
rating	float	Review rating	3.0 / 1.0 / 5.0

Column Name	Datatype	Description	Example
title	string	Title of the review	"Smells like gasoline! Going back!"
text	string	Full review text	"First & most offensive: they reek..."
images	array of objects	Each image has URLs + type; may be empty list	[{"small_image_url": "..."}]
asin	string	Product ASIN	"B083NRGZMM"
parent_asin	string	Parent ASIN	"B083NRGZMM"
user_id	string	Reviewer ID	"AFKZENTNBQ7A7V7UXW5JJI6UGRYQ"
timestamp	long / int64	Unix timestamp (ms)	1658185117948
helpful_vote	integer	Helpful vote count	0
verified_purchase	boolean	Whether Amazon confirmed purchase	true / false

4. Gift card

Column Name	Datatype	Description	Example
rating	float	Star rating	5.0
title	string	Review title	"Great gift"
text	string	Full review text	"Having Amazon money is always good."
images	array	List of image objects (empty here) []	
asin	string	Amazon product identifier	"B00IX1I3G6"
parent_asin	string	Variation parent ASIN	"B00IX1I3G6"
user_id	string	Reviewer ID	"AHZ6XMOLEWA67S3TX7IWEXXGWSOA"
timestamp	long / int64	Unix timestamp (ms)	1549866158332
helpful_vote	integer	Number of helpful votes	0 / 27
verified_purchase	boolean	Verified by Amazon	true / false

5.Sports and outdoors

Column Name	Data Type	Present In Description	
rating	Number (float/int)	All 3	Review star rating
title	String	All 3	Title of the review
text	String	All 3	Main review text
images	Array (list)	All 3	List of image URLs (empty in your samples)
asin	String	All 3	Amazon ASIN of product
parent_asin	String	All 3	Parent ASIN (variation parent)
user_id	String	All 3	Reviewer's anonymized ID
timestamp	Number (Unix ms)	All 3	Unix timestamp in milliseconds
helpful_vote	Number (int)	All 3	Helpful vote count
verified_purchase	Boolean	All 3	Whether purchase was verified
style	Object / Map	Only 1st	Variation metadata (size, color, etc.)

meta_Amazon_Fashion:

Column Name	Data Type	Description	Additional Info
main_category	String	Main category under which the Amazon product is listed.	Example: "AMAZON FASHION"
title	String	Full product title as shown on Amazon.	Useful for text analysis, NLP,

Column Name	Data Type	Description	Additional Info
			keyword extraction.
average_rating	Float	Average customer rating for the product.	Range: 1.0–5.0
rating_number	Integer	Number of customer ratings received.	Indicates popularity / reliability.
features	List	List of product features/bullet points.	Empty list in this record.
description	List	Detailed product description, usually paragraphs.	Empty list in this record.
price	Float / Null	Product price.	Null here because price is missing.
images	List of Objects	Contains multiple image versions for the product.	See breakdown below.
images[].thumb	String (URL)	Thumbnail image URL (low resolution).	Used for quick previews.
images[].large	String (URL)	Standard large image URL.	Normal resolution product image.
images[].variant	String	Image variation code such as MAIN, PT01, PT02, etc.	Distinguishes image type (main/side/angle).
images[].hi_res	String (URL)	High-resolution image URL.	Useful for zoomed views or downloads.
videos	List	Videos related to the product.	Empty for this item.
store	String	Seller or store name on Amazon.	Example: "GiveGift"
categories	List	Category/subcategory list.	Empty here.
details	Object	Detailed technical/packaging information.	See below.
details → Package Dimensions	String	Package size and weight.	"10.31 x 8.5 x 1.73 inches; 14.82 Ounces"
details → Item model number	String	Manufacturer item/model code.	Example: "DHE55PM21DH12"
details → Date First Available	String (Date)	The date the item was first published on Amazon.	Example: "February 12, 2021"
parent_asin	String	Parent ASIN (product group identifier).	Used for product variations.
bought_together	Null / Object	Items frequently bought together.	Null here (no data).

Meta_Gift_Card

Column Name	Data Type	Description	Additional Info
main_category	String	Main product category on Amazon.	"Gift Cards"
title	String	Full product title.	Gift Tag gift card.
average_rating	Float	Average user rating.	4.8
rating_number	Integer	Number of ratings.	1006 ratings
features	List of Strings	Bullet-point features.	8 features included.
description	List of Strings	Product description paragraphs.	Contains 1 description.
price	Float / Null	Product price.	Null (gift card value varies).
images	List of Objects	Contains image URLs in different qualities.	5 total images.
images[].thumb	String (URL)	Thumbnail image (small resolution).	For quick preview.
images[].large	String (URL)	Standard large image.	Medium resolution.
images[].variant	String	Image type/angle code (MAIN, PT01...).	Identifies the view.

Column Name	Data Type	Description	Additional Info
images[].hi_res	String (URL)	High-resolution version of image.	Highest image quality.
videos	List	Videos associated with product.	Empty list.
store	String	Seller's name.	"Amazon"
categories	List of Strings	Product category hierarchy.	["Gift Cards", "Gift Card Recipients", "For Him"]
details	Object	Additional metadata.	Contains 4 fields.
details → Package Dimensions	String	Item sizing and weight.	"5 x 3 x 0.1 inches; 0.63 Ounces"
details → Item model number	String	Model or variation identifier.	"Fixed"
details → Date First Available	String (Date)	Date item was listed on Amazon.	"August 29, 2017"
details → Manufacturer	String	Producing company.	"Amazon"
parent_asin	String	Parent ASIN for variation grouping.	"B06ZXTKYHN"
bought_together	Null/Object	Frequently bought together data.	Null

Meta_Electronics

Field Name	Data Type	Description
main_category	String	Top-level category to which the product belongs (e.g., <i>All Electronics</i>).
title	String	Product name as shown on Amazon.
average_rating	Float	Average customer rating (1.0 to 5.0 scale).
rating_number	Integer	Total number of customer ratings submitted.
features	List of Strings	Bullet-point product features or specifications; may be empty.
description	List of Strings	Long-form product description or marketing text.
price	Float / Null	Current price of the product; null if unavailable.
images	List of Objects	List of image records for the product.
└─ thumb	String (URL)	Thumbnail image URL.
└─ large	String (URL)	Large version of the image.
└─ variant	String	Image variant identifier (e.g., MAIN, PT01).
└─ hi_res	String / Null	High-resolution image URL if available.
videos	List	Video previews; empty if none available.
store	String	Name of the selling brand or store.
categories	List of Strings	Full Amazon category hierarchy for the product.
details	Object	Dictionary of detailed technical attributes.
parent_asin	String	Amazon Standard Identification Number for grouped parent products.
bought_together	String / Null	Items frequently bought together; null if not provided.

Product-Specific Details

Product 1: FS-1051 FATSHARK TELEPORTER V3 HEADSET

Field	Data Type	Description
details.Date First Available	Date (String)	The date the product was first listed on Amazon (August 2, 2014).
details.Manufacturer	String	Name of the manufacturer (<i>Fatshark</i>).

Category Path:

Electronics → Television & Video → Video Glasses

Images:
Contains 1 image (MAIN variant).

Product 2: Ce-H22B12-S1 4Kx2K HDMI 4-Port Splitter		
Field	Data Type	Description
details.Product Dimensions	String	Physical dimensions of the product: "0.83 x 4.17 x 2.05 inches".
details.Item Weight	String	Weight of the item: "5.3 ounces".
details.Item model number	String	Internal model identifier: "CE-H22B12-S1".
details.Is Discontinued By Manufacturer	String (Boolean-like)	Indicates whether the product has been discontinued.
details.Date First Available	Date (String)	When it was first listed on Amazon (June 3, 2015).
details.Manufacturer	String	Manufacturer name (<i>SIIG</i>).

meta_Clothing_Shoes_and_Jewelry

1. Top-Level Fields

Field	Example Value	Data Type	Explanation
main_category	"AMAZON FASHION"	String	High-level category.
title	"BALEAF Women's Long Sleeve..."	String	Product name/title.
average_rating	4.2	Float	Decimal rating.
rating_number	422	Integer	Count of ratings.
features	["90% Polyester...", ...]	List of Strings	Array describing key features.
description	[]	List	Empty list (no product description).
price	31.99	Float	Product price.
images	[{...}, {...}]	List of Objects	Collection of image metadata.
videos	[{title:..., url:...}]	List of Objects	Product videos and URLs.
store	"BALEAF"	String	Seller or brand.
categories	["Clothing...", "Women"...]	List of Strings	Category hierarchy.
details	{"Department": "womens", ...}	Object	Additional product attributes.
parent_asin	"B09X1MRDN6"	String	Amazon parent ASIN.
bought_together	null	Null	No paired products.

2. Inside images List		
Each image is an object containing:		
Field	Example	Data Type
thumb	URL string	String
large	URL string	String
variant	"MAIN", "PT01"	String
hi_res	URL string	String

3. Inside videos List		
Each video is an object :		
Field	Example	Data Type
title	"Women's UPF 50+..."	String
url	Amazon video URL	String
user_id	""	String (empty)

4. Inside details Object

Detail Key	Value	Data Type
Department	"womens"	String
Date First Available	"April 3, 2022"	String (Date in text form)

meta_Sports_and_Outdoors

1. Top-Level Fields

Field	Example Value	Data Type	Explanation
main_category	null	Null / Missing	No value provided.
title	"Sure-Grip Zombie Wheels Low 59mm 4 Pack"	String	Text-based product title.
average_rating	4.5	Float	Decimal numeric value.
rating_number	84	Integer	Whole number count of ratings.
features	["Pre-packaged..."]	List of Strings	Array of feature descriptions.
description	["All Zombie wheels..."]	List of Strings	Long multi-paragraph product description.
price	55.0	Float	Numeric product price.
images	[{thumb:..., large:...}]	List of Objects	Each object contains URLs and metadata about images.
videos	[]	List	Empty list (no videos).
store	"Sure-Grip"	String	Name of the seller/brand.
categories	["Sports & Outdoors", ...]	List of Strings	Hierarchical product categories.
details	{ ... }	Object / Dictionary	Key-value metadata about the product.
parent_asin	"B01HDXC8AG"	String	Amazon ASIN code.
bought_together	null	Null	No related products listed.

2. Inside images Field

Field	Example	Data Type
thumb	URL string	String
large	URL string	String
variant	"MAIN"	String
hi_res	URL string or null	String / Null

3. Inside details Object

Detail Key	Example Value	Data Type
Item Package Dimensions L x W x H	"4.49 x 4.49 x 1.5 inches"	String
Package Weight	"0.68 Kilograms"	String
Brand Name	"Sure-Grip"	String
Country of Origin	"United States"	String
Model Name	"zombie"	String
Color	"black 92A"	String
Material	"urethane aluminum"	String
Manufacturer	"Sure-Grip"	String
Part Number	"ZL92"	String
Style	"Zombie"	String

Detail Key	Example Value	Data Type
Size	"59mm"	String
Sport Type	"Roller Skating / Roller Blading"	String
Skill Level	"Professional"	String
Grip Type	"varies"	String
Best Sellers Rank	{"Sports & Outdoors":295175}	Object (String → Integer)
Date First Available	"March 16, 2015"	String (Date as text)

5] Key Performance Indicators (KPIs)

1. Return Prediction KPIs (Model Effectiveness)

- 1) Return Probability Score (RPS)
- 2) Model Accuracy / Precision / Recall / F1-Score
- 3) Top-K Risk Capture Rate

2. Product Quality KPIs

- 1) Product Return Risk Score (PRRS)
- 2) Rating–Return Discrepancy Index (RRDI)
- 3) Review Volatility Score

3. Seller / Retailer Performance KPIs

- 1) Seller Return Complaint Rate (SRCR)
- 2) Seller Quality Index (SQI)
- 3) Verified Purchase Trust Score

4. Recommendation System KPIs

- 1) Quality-Aware Recommendation Score (QARS)
- 2) Return-Risk Reduction Rate

5. Category-Level Risk KPIs

- 1) Category Return Risk Index (CRRi)
- 2) Category Sentiment Skewness

6. Business Impact KPIs

- 1) Estimated Return Cost Reduction
- 2) Customer Trust Index

6] ARCHITECTURE

